

Rainfall, Food Prices, and External Debt in Bangladesh

A Predictive and Clustering Analysis

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1. INTRODUCTION AND MOTIVATION

Bangladesh is among the world's most climate-exposed nations, where seasonal rainfall directly shapes agricultural output, food security, and macroeconomic stability. Individual linkages — climate to food security, and food prices to sovereign borrowing — are documented in the literature, yet no prior study has modelled the full pipeline **Rainfall → Food Prices → External Debt** within a unified machine-learning framework at the subnational level.

Auffhammer et al. (2012) showed rainfall shocks reduce rice yields across South Asia; Ivanic & Martin (2008) demonstrated that food price spikes drive external borrowing in import-dependent nations; Headey & Fan (2010) linked the 2007–08 food crisis to measurable debt increases; and Ahmed et al. (2020) confirmed that subnational rainfall anomalies predict Bangladesh market prices. This project quantifies and integrates those relationships across 1999–2024 using all eight administrative divisions.

Note: External debt is also driven by infrastructure finance, remittances, and RMG export cycles; results are framed as associational rather than strictly causal.

2. RESEARCH QUESTIONS

- **RQ1 (Supervised):** Do subnational rainfall levels and anomalies predict staple food prices (rice, wheat flour, palm oil) at the monthly division level?
- **RQ2 (Supervised):** Do aggregated national food price trends (lagged 3–12 months) associate with Bangladesh's external debt indicators at the annual level?
- **RQ3 (Clustering):** Do natural risk clusters emerge when division–year combinations are characterised by combined rainfall anomaly, food price deviation, and debt stress profiles?
- **RQ4 (Lag Analysis):** What is the optimal lag between rainfall anomaly and food price response, and between food price shock and debt-service pressure?

3. DATA SOURCES AND INTEGRATION

Five HDX / World Bank datasets are merged into a singel panel (~2,400 rows: 8 divisions × 25 years × 12 months, 15 columns). Merge key: PCODE (BD10–BD60) harmonised to admin1 division names; dekadal rainfall aggregated to monthly totals; annual debt values broadcast across 12 months; food prices averaged by commodity, division, and month.

Dataset	Source	Granularity	Key Columns Used
Subnational Rainfall (Full)	HDX / CHIRPS	adm_level=1, 8 divisions	rfh, rfh_avg, rfq (anomaly %), PCODE
Subnational Rainfall (5yr)	HDX / CHIRPS	adm_level=1 (recency layer)	r1h, r3h (1- and 3-month cumulative)
WFP Food Prices	HDX / WFP	Market → division–month avg	Rice (coarse), Wheat flour, Palm oil (USD/100 kg)
WFP Markets	HDX / WFP	Market metadata	market_id, admin1 (spatial join key)
External Debt (World Bank)	World Bank IDS	National → broadcast monthly	Debt total (DOD), Debt/GNI %, Debt service/exports %

4. METHODOLOGY

- ### 4.1 Preprocessing Pipeline
- Remove anomalous food price records (pre-1990 entries); impute sparse commodity gaps via division-level forward-fill after 2005.

- Construct lag features: **rfh_lag1–lag6** (1–6 month rainfall lags) for RQ1 & RQ4; national monthly food price index for RQ2.
- Normalise food prices to USD/100 kg; apply StandardScaler before all ML models.

4.2 Supervised Learning — Two Prediction Tasks

Three algorithms applied to both prediction tasks (Rainfall → Food Prices; Food Prices → Debt). **Evaluation:** 10-fold cross-validation; RMSE, MAE, and R^2 reported for every model.

Algorithm	Justification	Key Outputs
Linear Regression	Interpretable baseline; quantifies predictor direction and magnitude	Coefficients, RMSE, R^2
Decision Tree Regressor	Captures non-linear thresholds (e.g. drought cutoffs); visualised	Tree diagram, depth 3–8, min_samples_split tuned
Random Forest Regressor	Ensemble variance reduction; feature importance ranking	n_estimators 100–300, OOB score, permutation importance

4.3 Unsupervised Learning — K-Means Clustering

- **Unit of analysis:** Division × year vectors ($8 \times 25 = 200$). Features: annual rfq anomaly, rice price deviation from division baseline, ext_debt_pct_gni.
- **Optimal k:** Elbow method (WCSS) + Silhouette score; expected $k = 3$ –5. Hierarchical clustering used as robustness check.
- **Outputs:** PCA biplot, cluster centroid heatmap, division-level choropleth; qualitative labels (e.g. “Climate-Stressed High-Risk”, “Favourable”).

5. TECHNOLOGY STACK

Data ingestion & SQL joins: Apache Hadoop HDFS + Hive (dekadal aggregation, monthly averaging).

ML pipeline: Python 3 / Google Colab — pandas, numpy, scikit-learn, statsmodels, matplotlib, seaborn, folium.

Visualization: matplotlib (labeled figures, numbered axes), seaborn (heatmaps), folium (choropleth maps)

6. EXPECTED OUTPUTS AND CONTRIBUTION

- A clean, documented merged dataset (2,400+ rows, 15 columns) made openly available via GitHub.
- Cross-validated ML models with RMSE, R^2 , and feature importance for both prediction tasks; lag sensitivity chart for RQ4.
- K-Means risk clusters with qualitative interpretation and division-level choropleth visualisation.
- **Publication path:** With addition of Granger causality and ADF unit-root tests, the paper targets *Food Policy*, *Environmental Science & Policy*, or *Climate Risk Management* (Q2–Q3 journals).

7. LIMITATIONS

- **Scope:** External debt has many drivers beyond food prices (infrastructure loans, remittances, RMG exports); all results are framed as associations, not causal claims.
- **Data coverage:** Food price data is uneven before 2006 (Mymensingh & Rangpur available only from 2020); findings are stratified by data-rich vs. data-sparse divisions.
- **Clustering:** K-Means assumes spherical clusters and is sensitive to initialisation; hierarchical clustering is applied as a robustness check.

Approval